

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	15 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	State-wise Electricity Analysis	Display electricity consumption of all Indian states for 2019 and 2020
		Compare electricity consumption between states
		Display values on map visualization
FR-2	Year-wise Consumption Analysis	Compare electricity usage between 2019 and 2020
		Show monthly consumption trends
		Display yearly comparison charts
FR-3	Regional Analysis	Display electricity consumption by region
		Compare consumption among North, South, East, West, and Central regions
FR-4	Quarterly & Monthly Analysis	Show quarter-wise electricity consumption
		Display month-wise trends
FR-5	Lockdown Impact Analysis	Compare electricity usage before and during COVID-19 lockdown
		Analyze recovery after lockdown
FR-6	Top & Bottom State Analysis	Display Top-N electricity consuming states
		Display Bottom-N electricity consuming states
FR-7	Dashboard Visualization	Show interactive dashboards
		Display filters for Year and State

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Allow dashboard navigation
FR-8	Story Presentation	Display insights using Tableau Story
		Navigate between story points
FR-9	Data Refresh	Refresh electricity dataset
		Update visualizations automatically
FR-10	Publish Dashboard	Publish dashboard to Tableau Public
		Generate shareable public link

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be easy to understand and navigate.
NFR-2	Performance	Visualizations should load within a few seconds.
NFR-3	Reliability	Dashboard should display accurate electricity consumption data.
NFR-4	Availability	Tableau Public dashboard should be accessible online 24x7.
NFR-5	Scalability	System should support additional years and new datasets in the future.
NFR-6	Maintainability	Dashboard should be easy to update when new data becomes available.
NFR-7	Data Accuracy	Visualizations should represent the dataset correctly without inconsistencies.
NFR-8	Compatibility	Dashboard should work across modern web browsers and desktop devices.
NFR-9	Security	Only authorized users should modify the Tableau workbook; viewers have read-only access on Tableau Public.
NFR-10	Portability	Workbook should be portable and shareable through Tableau Public.